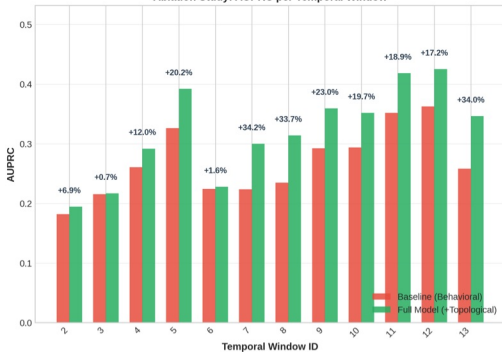
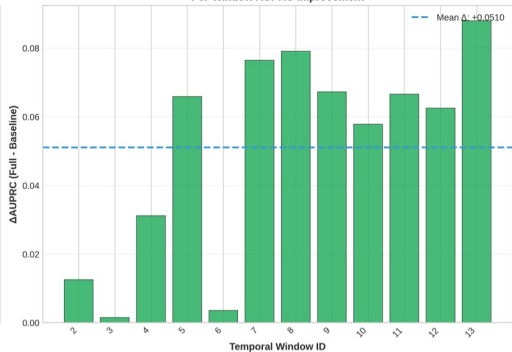


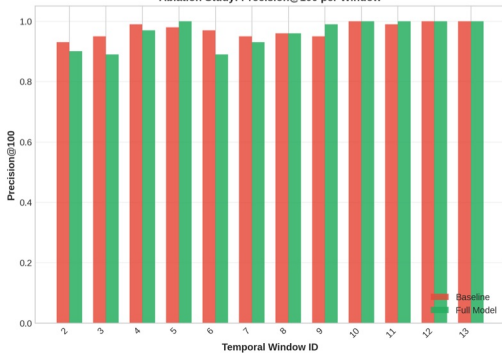
Ablation Study: AUPRC per Temporal Window



Per-Window AUPRC Improvement



Ablation Study: Precision@100 per Window



ABLATION STUDY: STATISTICAL SUMMARY

AUPRC (Area Under Precision-Recall Curve)

Mean Δ : +0.0510 $\pm$ 0.0306
t-statistic: -5.7817
p-value: 1.2253e-04
Significant: $\checkmark$ YES ($\alpha=0.05$)

ROC-AUC

Mean Δ : +0.0268 $\pm$ 0.0048
p-value: 6.8686e-10

Precision@100

Mean Δ : -0.0117 $\pm$ 0.0333
p-value: 2.4973e-01

CONCLUSION:

Topological features provide STATISTICALLY SIGNIFICANT improvement in fraud detection performance.